

Lab 10: Designing RF Filters

ETE 4371

Sarah Boyd

Joanna Escobar

Mariana Rivera

Introduction

In this lab, we will be going over RF filters and amplifiers. We have had some experience designing filters and amplifiers in earlier classes, such as ETE 3501 Control Systems, but they were for systems with lower frequencies. This lab will focus more on high frequency filtering applications. This is an important concept for data transmission, because if a high data rate, high frequency transmission is corrupted with low frequency noise, we need a way to remove the noise from the signal. There are many ways to create a filter, but we will be focusing on using a guide from our textbook in this demo.

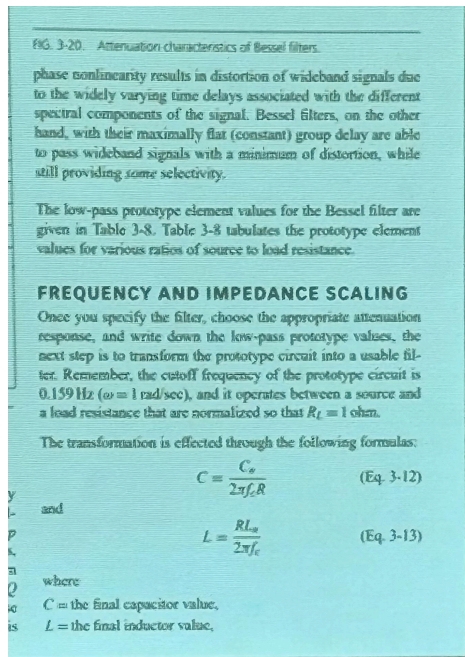
Demo

Filters

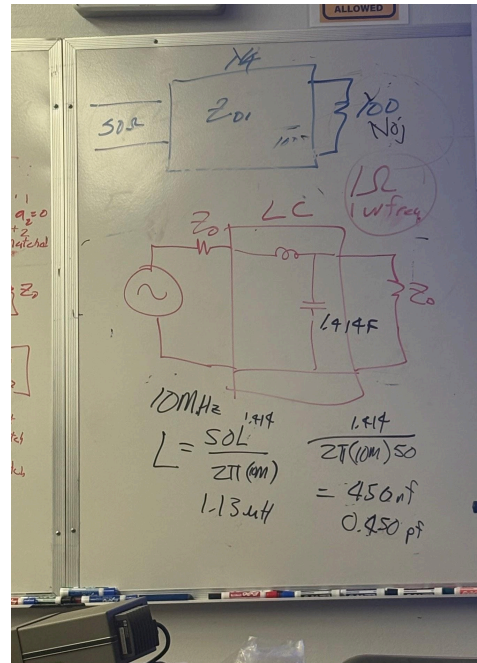
RF filters are different than other filters because the characteristic impedance (Z_0) in front and on the load need to get matched. In addition, there are also no resistors in this system. The values that we decided to design for were the following:

- $R=50$ ohms no change
- Design for 10 Mhz

In order to calculate for value, we used the equations found in our textbook "Electronic Transmission Technology" by William Sinnema.

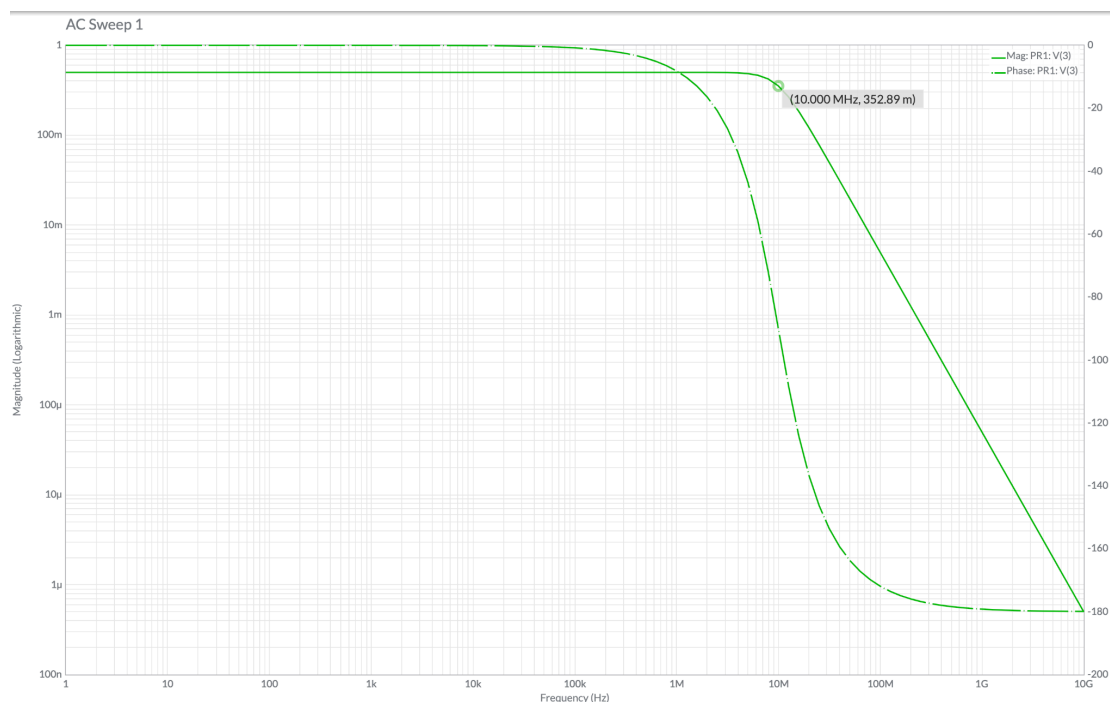


IMG. 1: Equations from the textbook.

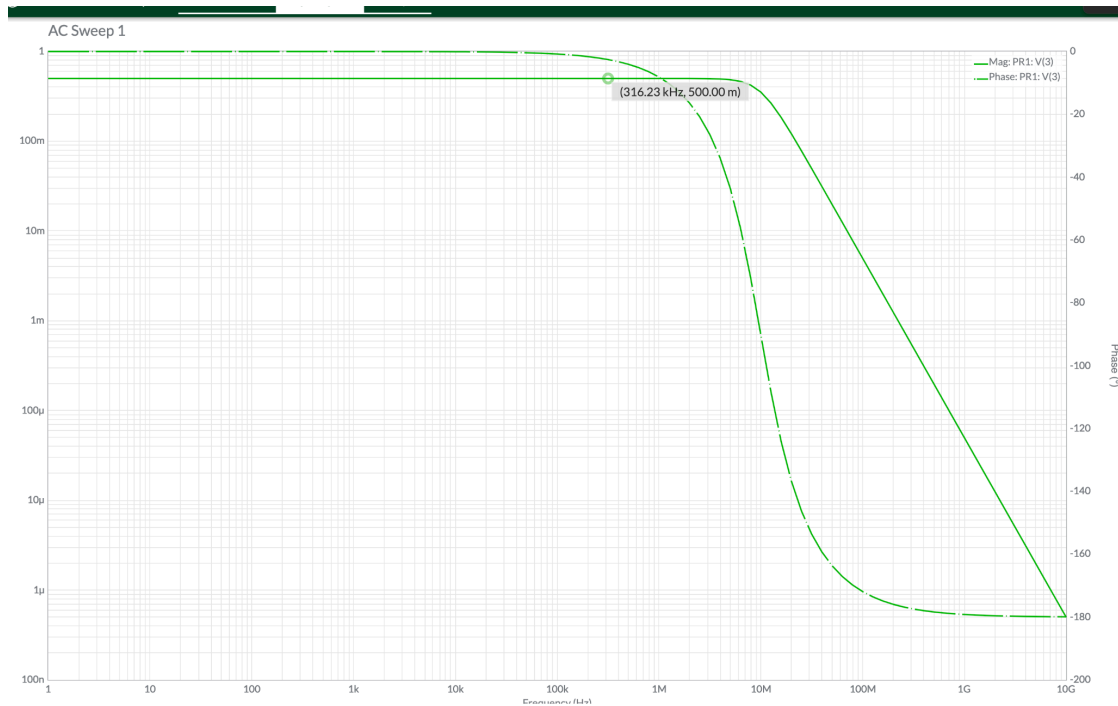


IMG. 2: Worked out diagram for the filter.

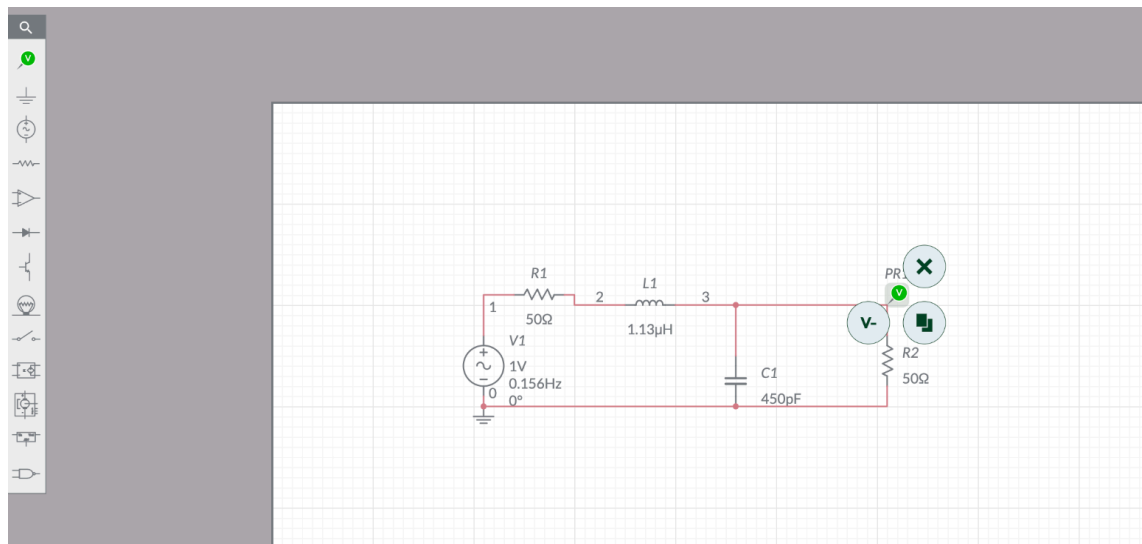
Then, we take those values we got and adjust our circuit with the values we want:



IMG. 3: Roll off of the filter at 10MHz.



IMG. 4: Filter at 316.23 kHz.



IMG. 5: Circuit diagram of the filter.

RF filters will always have 50 ohm resistors in its circuit diagram. This is in contrast to other filters that use resistors in the kOhms to tens of kOhms range.

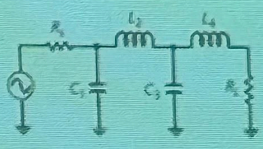
In this case, we used a scaling factors table for all filters. This table is used to determine the number of and value of the inductive and capacitive components. This is done in order to allow

for quicker design of the filter than if we were to do all of the math by hand. This example is a Butterworth filter, which is commonly used in control systems because of the flat filter and slow roll-off. In a lower frequency system (like many physical systems), this is acceptable. In an RF system, a slow roll-off may not be acceptable. This is where the Chebyshev filter comes in. The Chebyshev filter has a much steeper roll-off than a Butterworth filter. However, the trade off for this is that there is a ripple on the passband. In systems where you don't want variation in signal levels, this can cause issues. In communications though, a little ripple in the dB measurement of the filtered signal is not a big deal. This, paired with the sharp cut off makes the Chebyshev filter a popular filter to use in RF and Microwave systems.

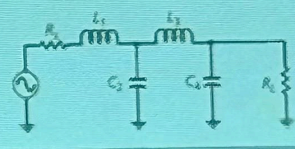
Downloads/RF_Circuit_Design_....(CHAPTER_3_Filter_Design).pdf

Mail - Gerald K. Ho... Customize Links Free Hotmail Windows Marketplace Imported From IE

A 3 Ask Copilot 16 of 26



n	R_H/R_L	C_1	ϵ_3	C_3	L_4
2	3.000	0.572	3.332		
	4.000	0.365	4.600		
	8.000	0.157	9.658		
	∞	1.213	1.109		
3	1.000	2.216	1.088	2.216	
	0.500	4.431	0.817	2.216	
	0.333	6.647	0.726	2.216	
	0.250	8.862	0.680	2.216	
	0.125	17.725	0.612	2.216	
	∞	1.652	1.460	1.108	
4	3.000	0.653	4.411	0.814	2.535
	4.000	0.452	7.083	0.612	2.848
	8.000	0.209	17.164	0.428	3.281
	∞	1.350	2.010	1.488	1.106
n	R_L/R_S	L_1	C_2	L_3	C_4



EXAMPLE 3-5

Scale the low-pass prototype values of Fig. 3-19 (Example 3-4) to a cutoff frequency of 50 MHz and a load resistance of 250 ohms.

Solution

Use Equations 3-12 and 3-13 to scale each component as follows:

$$C_1 = \frac{3.545}{2\pi(50 \times 10^6)(250)} = 45 \text{ pF}$$

$$C_3 = \frac{9.127}{2\pi(50 \times 10^6)(250)} = 116 \text{ pF}$$

$$C_5 = \frac{7.889}{2\pi(50 \times 10^6)(250)} = 100 \text{ pF}$$

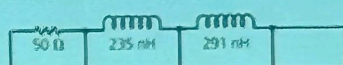
$$L_2 = \frac{(250)(0.295)}{2\pi(50 \times 10^6)} = 235 \text{ nH}$$

$$L_4 = \frac{(250)(0.366)}{2\pi(50 \times 10^6)} = 291 \text{ nH}$$

The source resistance is scaled by multiplying its normalized value by the final value of the load resistor.

$$R_{\text{final}} = 0.2(250) = 50 \text{ ohms}$$

The final circuit appears in Fig. 3-21.

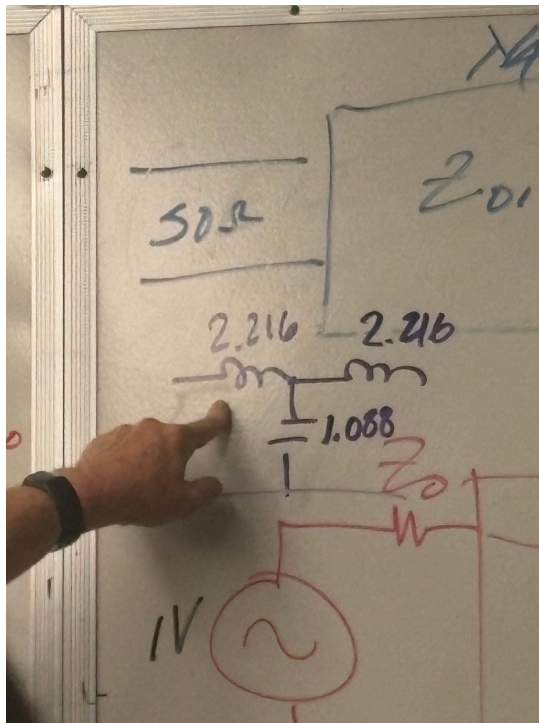


IMG. 6: Filter design table.

For the second filter, we were walked through how to design a 1 dB Chebyshev filter. There are a few steps to determining the inductive and capacitive values in the circuit. They include the following.

1. Look for the filter order in the first column.
2. Look down the chart for CLC or bottom up for LCLC.

In this next example, we will be making a third order LCL filter. We can get the first L value using the table in IMG. 6. In this case, it is 2.216 because our source and load resistor are matched. Going along that row, we can see that the C value will be 1.088, and the second L will also be 2.216.



IMG. 7: Third order filter design.

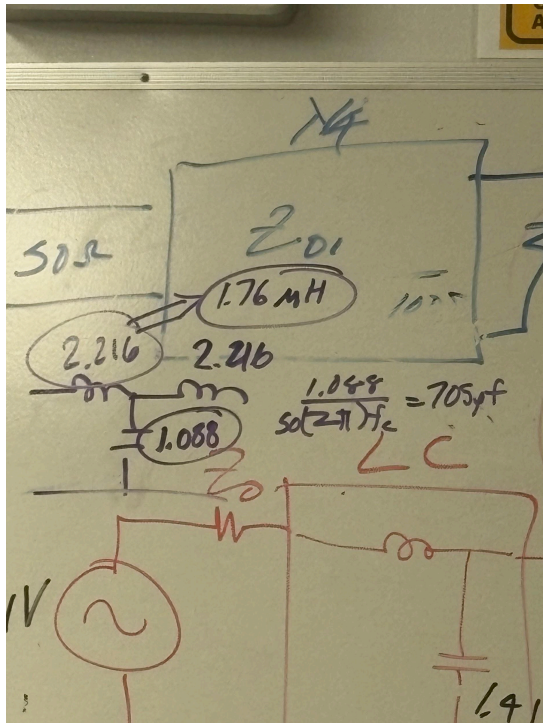
Now that we know the most significant digits, we need to know what the scale is. We will figure out the scaling factor by modifying the formulas seen in IMG. 6 under “Example 3-5”.

$$L_1 = \frac{RL_1}{2\pi f_c} = \frac{(50)(2.216)}{2\pi(10\text{MHz})} = 1.76\mu\text{H}$$

$$C_1 = \frac{C_1}{R_L * 2\pi * f_c} = \frac{1.088}{(50)(2\pi)(10\text{MHz})} = 705\text{pF}$$

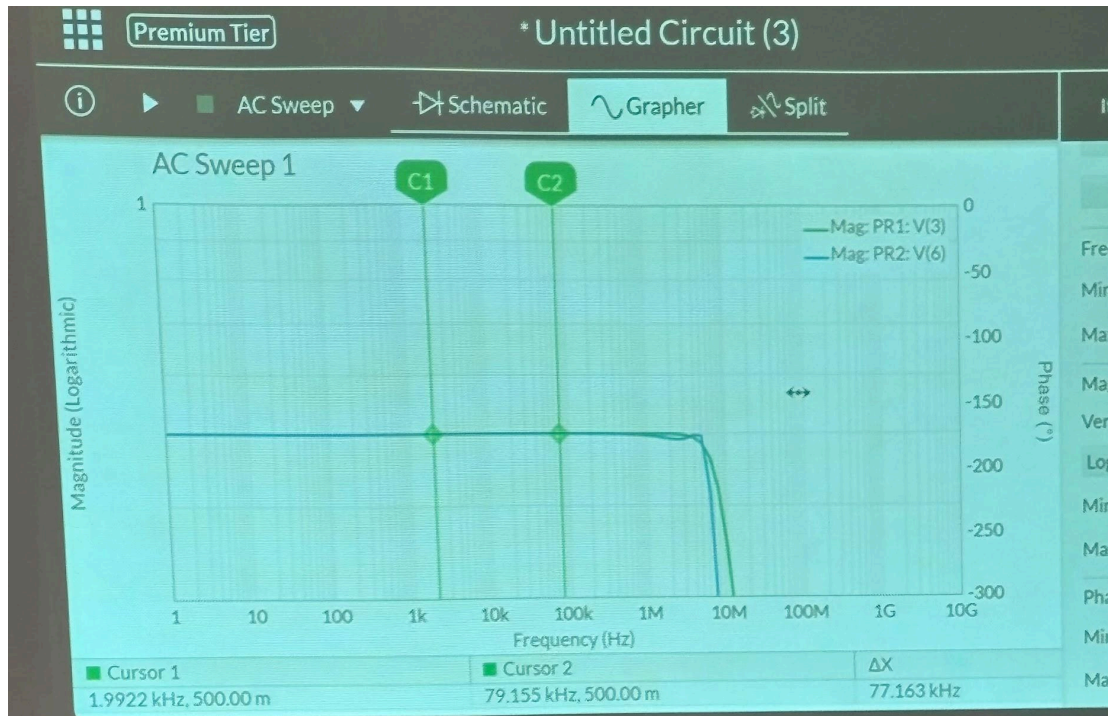
$$L_2 = \frac{RL_2}{2\pi f_c} = \frac{(50)(2.216)}{2\pi(10\text{MHz})} = 1.76\mu\text{H}$$

After scaling using the formulas, we know that the inductors have a value of 1.76 uH, and the capacitors have a value of 705 pF.



IMG. 8: Scaling Factor Calculations.

Now we adjust the filter with the values we just calculated. We can see that the roll off is much faster than with the Butterworth filter, as we would expect. We can also zoom in to see the ripple at the edge of the passband, and the frequency response of the circuit.



IMG. 8: Chebyshev filter roll off and frequency response.

We can create any filter using either this table or a similar filter table, as long as we know the correct order to read the L and C values. For instance, for a High Pass filter, the Ls and Cs are switched as compared to how we used them in this demo. Band pass filters require their own chart, because they are essentially both high pass and low pass filters operating at the same time. All of these filter tables are in our green textbook.

Amplifiers

RF amplifiers and the amplifiers we've worked with previously have some important differences. RF amplifiers have input, output and power connections, unlike amplifiers at the component level, where you have to solder on power connections and have most of their signals sealed inside the "black box".

One challenge we learned about was that the amplifier's signal out frequency spectrum can be brought up to the point where it will damage the network analyzer. The network analyzer is sensitive, because it has to be able to detect signals with very high frequencies. In most cases, the network analyzer will shut itself off or reduce the amplitude and current of the signal it is sending to the amplifier before any damage occurs, but it is still good practice to be aware of any possible dangers.

A way to safely test the amplifier is to connect an attenuator to the output of the amp. This attenuator will reduce the signal so the two components cancel each other out. This is safer for the network analyzer, while still allowing for you to be able to record data from the amplifier. In addition, attenuators do not have a frequency response, so we do not have to worry about it affecting the recorded output of the network analyzer.

Coaxial Amplifier
ZFL-1000H+
ZFL-1000H

50Ω Medium Power 10 to 1000 MHz

Features

- wideband, 10 to 1000 MHz
- low noise, 5 dB typ.
- high IP3, +33 dBm typ.
- protected by US Patent, 6,943,629

Applications

- cellular
- VHF/UHF
- test equipment

Electrical Specifications

Parameter	Min.	Typ.	Max.	Units
Frequency Range	10	—	1000	MHz
Gain	28	—	—	dB
Gain Flatness	—	—	+1.0	dB
Output Power at 1dB compression	+20	—	—	dBm
Noise Figure	—	5.0	—	dB
Output third order intercept point	—	+33	—	dBm
Input VSWR	—	2.0	—	:1
Output VSWR	—	2.0	—	:1
DC Supply Voltage	—	15	—	V
Quiescent Current	—	—	100	mA

Case Style: 5508

Connection	Model	Price	Qty.
SMA	ZFL-1000H(+)	\$219.00 ea.	(1-6)
BRACKET (OPTION "B")		\$2.50	(1-)

To order without heat sink see data sheet for ZFL-1000HX(+)

RoHS Compliant
 The eSata identifies RoHS Compliance. See our web site for RoHS Compliance methodologies and qualifications.

IMG. 9: Datasheet of the amplifier used in the demo.

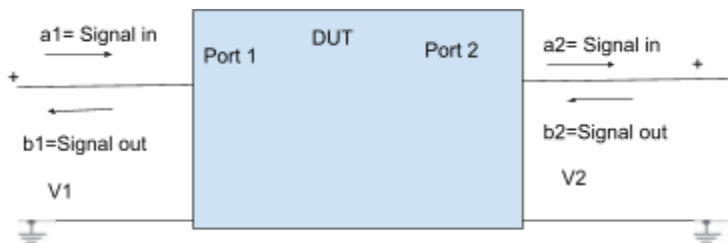
In class, we were able to see that one of the outputs has a gain of almost 28dB, so the power gain, or $\frac{P_{out}}{P_{in}}$, is $10^{\frac{28}{10}} = 631$ linear power gain. Therefore, the output power was about 631 times greater than the input power. This is close to 30dB, which would represent a 1000:1 power gain. We also saw that another had a gain of 10 dB. 3dB is a good gain, a doubling of the input signal as seen in table 1. The relationship between dB change and linear power change can be summarized as table 1.

Linear change	dB change
x2	+3 dB
x10	+10 dB
/2	-3 dB
/10	-10 dB

Table 1. *Converting between linear power ratio to dB*

We also looked at the amplifiers behavior over a range of frequencies. Unlike ideal amplifiers, real RF amplifiers do not have a perfect, constant gain at every frequency. Instead the gain can vary depending on frequency, input power, output power and how well the amplifier is matched to the rest of the 50Ω system. Because of this the VNA plots are useful due to how they allow us to see the amplifier's performance across a variety of frequencies as opposed to just one singular frequency. RF devices do not do well with operational amplifiers, because there is always that variation in gain depending on power. More voltage in the system results in more power, and the impacts can be observed by looking at the S parameter S21.

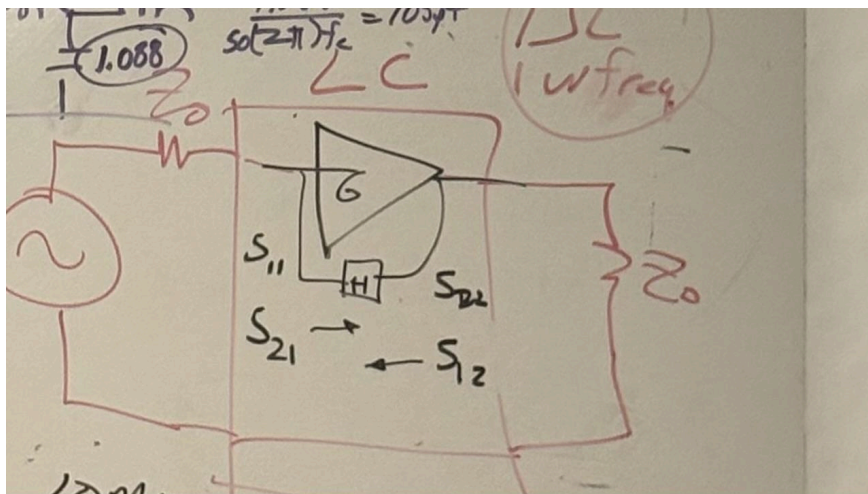
After connecting the amplifier, we saw that S11 and S22 on the VNA are matched impedances on the Smith chart, as expected. This matches expectations, as for an amplifier S11 and S22 should be a good match (return loss is a high negative value of -20dB or below). It is seen to be -40dB for S11, which indicates a great match and low reflection. We saw that S21 was greater than 0db- this track, because S21 is essentially out/in, or the transfer function, or the gain/loss. Therefore it is the only parameter that can be greater than 0dB, in the case of an amplifier. Additionally, we saw that the VSWR was 2:1. This indicates that the amplifier isn't perfect- which is to be expected- and is hard to match around a large variety of frequencies.



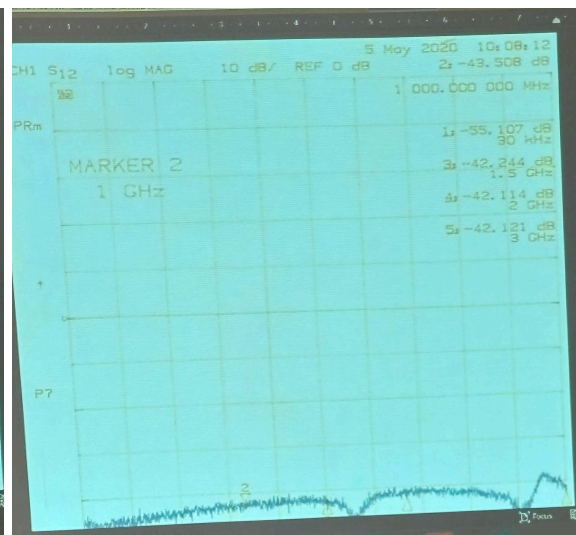
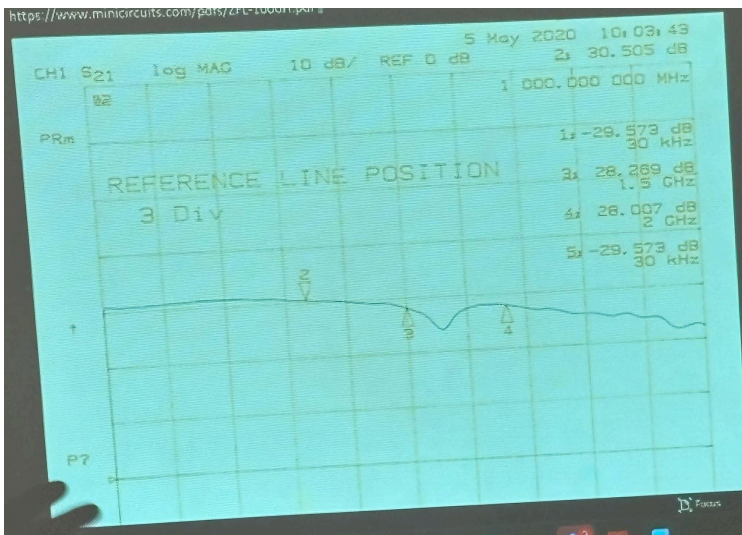
IMG 10. *Two Port Device Under Test*

Parameter	Explanation	Ratio	Ideal	Format
S_{11}	$S_{11} = \frac{\text{power reflected}}{\text{power incident}} = \text{input return/signal loss.}$ Essentially, S_{11} tells us how much the signal sent into port 1 reflects back out of port 1, or how well the device matches. A smaller value is a better match and a smaller reflection.	$\frac{b1}{a1}$	More negative dB is better, or closer to the center on a smith chart	Smith and Log
S_{12}	S_{12} is the reverse gain/ loss, or how much of the signal sent into port 2 comes out of port 1.	$\frac{b1}{a2}$	Smaller is better	Log
S_{21}	$S_{21} = \frac{\text{out}}{\text{in}} = \text{transfer function.}$ S_{21} is the forward gain/loss or insertion loss, and tells us the gain or loss going forward, and how much less signal the output has compared to the input. S_{21} is for a two port system. This is the only parameter that can be greater than 0dB, in the case of an amplifier.	$\frac{b2}{a1}$	Closer to 0db is good for most systems	Log
S_{22}	S_{22} is the output loss. It tells us how much of the signal sent into port 2 reflects back out of port 2, and how well output is matched.	$\frac{b2}{a2}$	More negative dB is better, or closer to the center on a smith chart	Smith

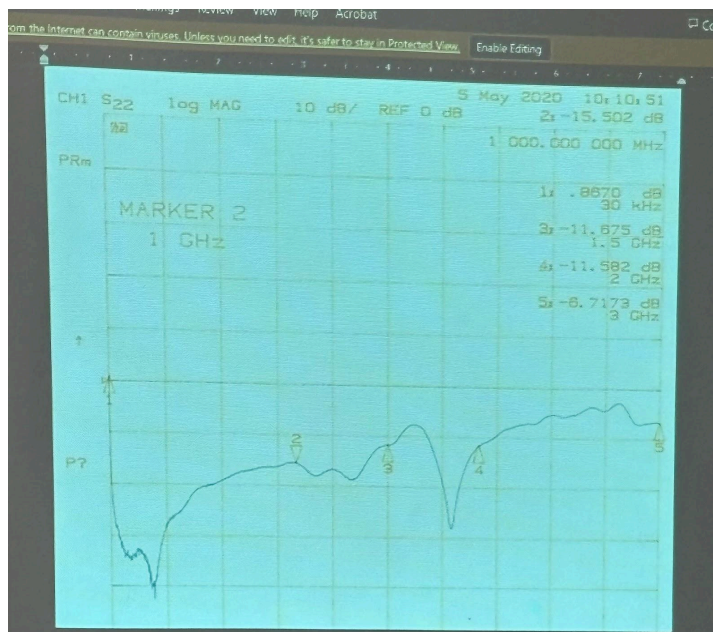
Table 2. S parameters



IMG 11. Circuit Diagram of the RF Amplifier.



IMG. 12: VNA Mag-Log output of the amplifier S21. **IMG. 13:** VNA Mag-Log output of S12.



IMG. 14: VNA Mag-Log output of S22.

Conclusion

Overall, the amplifier measurements showed that the device provided forward gain as expected while also demonstrating real world limitations such as gain variation and non perfect matching. The S parameter plots helped show the amplifiers behavior more clearly by separating input reflection, output reflection, forward gain and reverse gain into S11 S12 S21 and S22. This made it easier to understand why RF circuits - filters, amplifiers, attenuators, and DUT- are tested as black box systems rather than individual components. At high frequencies it is hard to analyze every aspect of every component, so measuring how the device behaves at its ports is much easier. This relates to the design of RF filters, as when designing them we care whether the desired frequencies pass through, unwanted are attenuated and whether the filter is properly matched so signals are not reflected back.